Software Specifications for Restaurant Management Version- 1.0

1. Introduction

1.1 Purpose

The purpose of this Software Specification is to describe what the Restaurant Management System should do and how it should function. It helps all stakeholders clearly understand the system requirements. The document defines both functional features such as order, inventory, customer, staff, and report management, and non-functional requirements like usability, performance, security, and scalability, ensuring the software meets business needs and works reliably on web and mobile platforms.

1.2 Target Audience0

1. **Restaurant Owners & Stakeholders**
    - Understand business value and make decisions.
2. **Managers & Supervisors**
    - Oversee operations, inventory, and staff.
3. **Kitchen & Front-of-House Staff**
    - Use order, reservation, and KDS features.
4. **Developers & Engineers**
    - Build, integrate, and maintain the system.
5. **UI/UX Designers**
    - Design user-friendly interfaces.
6. **QA & Testing Teams**
    - Verify system functionality and performance.
7. **IT/Admin Teams**
    - Handle deployment, security, and support.
8. **Third-Party Partners**
    - Integrate payment, delivery, and POS services.

1.3 Scope

In Scope

- **Order Management:** Online ordering, table reservations, real-time order tracking, KDS, and POS integration**.**
- **Inventory Management:** Real-time stock tracking, low-stock alerts, supplier management, and recipe costing.

- **Customer Management:** Customer profiles, order history, loyalty programs, and feedback management.
- **Reporting & Analytics:** Sales, inventory, and customer behavior reports with dashboards.
- **Staff Management:** Employee roles, scheduling, attendance tracking, and access control.
- **UI/UX:** Responsive web and mobile interfaces with role-based dashboards.
- **Integrations:** Payment gateways, delivery services, third-party tools, and APIs.
- **Security & Compliance:** Authentication, data encryption, regulatory compliance, and backups.
- **Performance & Scalability:** High availability, scalable architecture, multi-branch support.
- **Deployment & Maintenance:** SaaS deployment, monitoring, updates, and customer support.

Out-of-Scope:

- **Customer Mobile Apps:** No Android or iOS apps for customer ordering or reservations.
- **AI/ML Features**: No forecasting, recommendations, or intelligent analytics.
- **Delivery Operations: Customer Mobile Apps:** No Android or iOS apps for customer ordering or reservations.
- **AI/ML Features:** No forecasting, recommendations, or intelligent analytics.
- **Delivery Operations:** No delivery fleet or rider management.
- **Hardware:** No procurement or maintenance of POS, printers, or kitchen devices.
- **International Compliance:** No multi-country tax, currency, or legal support beyond GST.
- **Advanced CRM & Marketing:** No automated campaigns or advanced customer segmentation.
- **Security Certifications:**NoPCI-DSS, ISO, or enterprise-level certifications.
- **ERP Integrations:** No integration with enterprise ERP or finance systems.
- **Offline Mode:** No full offline or offline-first functionality.
- **Data Migration:** No legacy or historical data migration.
- **Custom Enhancements:** No features beyond approved SRD requirements.
- **Business Process Changes:** No redesign of restaurant workflows or staff training.

1.4 Definitions and Acronyms

- **SRD (Software Requirements Document):** A document that defines system requirements and scope.
- **RMS (Restaurant Management System):** Software used to manage restaurant operations.
- **POS (Point of Sale):** System used for billing and payment processing.
- **KDS (Kitchen Display System):** Digital screen used in kitchens to display orders.
- **CRM (Customer Relationship Management):** System for managing customer data and interactions.
- **UI (User Interface):** Visual layout through which users interact with the system.
- **UX (User Experience):** Overall experience of the user while using the system.
- **API (Application Programming Interface):** Interface that allows systems to communicate.
- **SaaS (Software as a Service):** Cloud-based software delivered via subscription.
- **KPI (Key Performance Indicator):** Metric used to measure system or business performance.
- **GST (Goods and Services Tax):** Tax applied to goods and services transactions.
- **UAT (User Acceptance Testing):** Final testing done by users before deployment.

2. Overview

2.1 System Description

1. **Presentation Layer (UI Layer):**
   Web and mobile user interfaces that allow customers, staff, and administrators to interact with the system.

2. **Application Layer (Business Logic Layer):**
   Handles core business logic such as order processing, inventory management, billing, user authentication, and integrations**.**

3. **Data Layer (Database Layer):**
   Manages secure storage and retrieval of data including orders, customers, inventory, staff, and reports.

2.2 User Needs

The system is designed to meet the needs of two primary user groups:

- **Restaurant Users (Internal Users):**
  Includes restaurant owners, managers, cashiers, kitchen staff, and service staff who use the system to manage orders, billing, inventory, staff, and reports.

- **Customers (External Users):**
  Includes customers who interact with the system for placing orders, making reservations, tracking orders, and providing feedback.

2.3 Assumptions and Dependencies

Assumptions

1. **Technology Readiness**
   - Restaurants have reliable internet connectivity to support cloud-based operations.
   - Users (staff and customers) have access to modern web browsers or smartphones.
   - Existing POS systems support standard APIs or integration mechanisms**.**
2. **User Adoption**

- Restaurant staff are willing to use digital systems for orders, inventory, scheduling, and reporting.
- Customers are comfortable placing orders and reservations online via web or mobile apps.
- Management will actively use reports and analytics for decision-making.

3. **Operational Processes**
   - Restaurants follow standardized workflows for ordering, inventory, and staff management.
   - Inventory items, recipes, and pricing data are accurately maintained by restaurant staff.
   - Loyalty programs and customer feedback processes are clearly defined by the business.
4. **Data Accuracy**
   - Initial data migration (menu, inventory, customer data) is accurate and complete.
   - Staff regularly update inventory, attendance, and order status to ensure real-time accuracy.
5. **Security & Compliance**
   - Restaurants agree to follow recommended security practices (strong passwords, role-based access).
   - Users consent to data collection in compliance with applicable regulations (GDPR, CCPA).
6. **Scalability Expectations**
   - System usage will grow gradually, allowing infrastructure to scale without abrupt spikes.
   - Performance benchmarks are acceptable for both small and multi-branch restaurant operations.

Dependencies

1. **Third-Party Services**
   - Payment gateway providers (e.g., cards, wallets) for online payments.
   - Delivery service APIs for order fulfillment and tracking.
   - SMS, email, and push notification services for customer and staff communication.
2. **POS System Integration**
   - Availability and stability of APIs from existing POS vendors.
   - Compatibility between the Restaurant Management System and POS data formats.
3. **Infrastructure & Hosting**

- Cloud service providers for hosting, storage, backups, and scalability.
- High availability of servers to meet performance and uptime **requirements.**

4. **Mobile & Web Platforms**
   - App store approvals (Google Play / Apple App Store) for mobile applications.
   - Continuous browser compatibility updates for web platforms.
5. **Security & Compliance Frameworks**
   - Encryption libraries and authentication services.
   - Regulatory updates impacting data privacy and payment processing.
6. **Operational Support**
   - Availability of trained customer support staff for 24/7 assistance.
   - Ongoing maintenance, bug fixes, and feature upgrades from the development team.
7. **Business Inputs**
   - Timely input from restaurant owners for configuration, pricing plans, and feature prioritization.
   - Clear subscription models and billing rules for SaaS implementation.

3. Requirements

3.1 Functional Requirements (FR)

1.1User Managment

- The system shall allow users to register and log in securely.
- The system shall support role-based access (Admin, Manager, Staff, Customer).
- The system shall allow admins to add, edit, or deactivate user accounts.

1.2 Order Management

- The system shall allow customers to place online food orders.
- The system shall allow staff to create and modify in-restaurant orders.
- The system shall display real-time order status (Placed, Preparing, Ready, Delivered).
- The system shall route orders to the Kitchen Display System (KDS).

1.3 Table Reservation

- The system shall allow customers to reserve tables online.
- The system shall allow managers to approve, modify, or cancel reservations.
- The system shall maintain a waitlist for fully booked time slots.

1.4 Menu Management

- The system shall allow admins to add, update, or remove menu items.
- The system shall display item descriptions, prices, and availability.
- The system shall support category-wise menu organization.

1.5 Inventory Management

- The system shall track inventory levels in real time.
- The system shall generate alerts for low-stock items.
- The system shall allow purchase order creation for suppliers.
- The system shall calculate recipe costing.

1.6 Payment Processing

- The system shall support multiple payment methods (Cash, Card, Online Wallets).
- The system shall generate digital invoices and receipts.
- The system shall integrate with third-party payment gateways.

1.7 Customer Relationship Management (CRM)

- The system shall store customer profiles and order history.
- The system shall support loyalty programs and reward points.
- The system shall collect customer feedback and ratings.

1.8 Reporting & Analytics

- The system shall generate daily, weekly, and monthly sales reports.
- The system shall provide inventory usage and wastage reports.
- The system shall analyze customer behavior and preferences.

1.9 Staff Management

- The system shall allow managers to create employee schedules.
- The system shall track staff attendance.
- The system shall manage role-based permissions**.**

2. Non-Functional Requirements (NFR)

Non-functional requirements describe how the system should perform.

2.1 Performance

- The system shall respond to user actions within 2 seconds under normal load.
- The system shall support concurrent users without performance degradation.

2.2 Security

- The system shall encrypt sensitive data (passwords, payment details).
- The system shall implement secure authentication and authorization.
- The system shall comply with data protection regulations (GDPR, CCPA).

2.3 Usability

- The system shall provide a user-friendly and intuitive interface.
- The system shall be accessible on web and mobile devices.
- The system shall require minimal training for staff users.

2.4 Reliability

- The system shall have an uptime of 99.9%.
- The system shall perform regular automated data backups.
- The system shall support disaster recovery mechanisms.

2.5 Scalability

- The system shall support multiple restaurant branches.
- The system shall handle increasing numbers of users and transactions.
- The system shall allow feature expansion without major redesign.

2.6 Compatibility

- The system shall be compatible with major browsers (Chrome, Firefox, Edge).
- The system shall support Android and iOS platforms.
- The system shall integrate with third-party POS and payment systems.

2.7 Maintainability

- The system shall allow easy bug fixes and feature updates.
- The system shall follow modular and documented code architecture.
- The system shall support regular software updates.

2.8 Availability & Support

- The system shall provide 24/7 customer support options.
- The system shall include a help section and FAQs.

4. Validation and Acceptance Criteria

Validation Criteria

- The system shall support online ordering, table reservations, and real-time order tracking on web and mobile platforms.
- The system shall correctly integrate with POS systems, payment gateways, and third-party services.
- Inventory levels shall update in real time, with automatic low-stock alerts.
- Customer data (profiles, orders, loyalty points) shall be accurately stored and retrievable.
- Reports (sales, inventory, customer behavior) shall generate accurate and consistent results.
- Role-based access control shall restrict features according to user permissions.
- All sensitive data shall be encrypted, backed up, and compliant with GDPR/CCPA.
- The system shall perform reliably under peak load conditions and scale as usage increases.

Acceptance Criteria

- Users can successfully place, track, and manage orders without errors.
- Table reservations can be created, modified, and canceled with proper time-slot handling.
- Kitchen staff receive orders instantly via the Kitchen Display System (KDS).
- Inventory alerts trigger when stock falls below defined thresholds.
- Loyalty points and rewards are calculated and applied correctly.
- Reports load within acceptable time limits and reflect accurate data.
- The UI is intuitive, responsive, and mobile-friendly.
- The system passes security, performance, and integration testing with no critical defects.
- Customer support features and SaaS subscription plans are accessible and functional.

5. Additional Sections

Glossary

Risk and Mitigation Strategies

- **Risk:** Potential events or conditions that could negatively impact system performance, security, or business operations.
- **Mitigation Strategy:** Planned actions taken to reduce the likelihood or impact of identified risks.
- **Operational Risk:** Risks related to system failures, human errors, or process inefficiencies.
- **Security Risk:** Threats such as data breaches, unauthorized access, or cyberattacks.
- **Technical Risk:** Risks arising from software bugs, integration issues, or scalability limitations.

Future Considerations

- **Scalability Enhancements:** Ability to support more users, locations, and higher transaction volumes.
- **Feature Expansion:** Addition of advanced analytics, AI-based recommendations, or automation tools.
- **Technology Upgrades:** Adoption of new technologies to improve performance, security, and usability.
- **Market Adaptability:** Adjusting the system to meet changing customer demands and industry trends.
- **Global Expansion:** Support for multiple languages, currencies, and regional regulations.

Appendix

- **Supporting Documents:** Additional materials such as diagrams, specifications, or reference data.
- **Technical Details:** Extended technical descriptions not included in the main sections.
- **Assumptions:** Key assumptions made during system design and planning.
- **References:** External sources, standards, or tools used in system development.
- **Acronyms & Abbreviations:** List of shortened terms used throughout the document.